An Agentic AI Approach with humans in the loop: Exploring the case of climate education through evidence synthesis - Appendices

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Contents

1 Appendix A: Examples of Climate Literacy Impact Anchor statements

```
clii_anchors = {
    "Cognitive": [
        "knowledge about climate change, e.g. Knowing that the
           melting of permafrost affects the Earth's climate or
           understanding the role of roof gardens in reducing
           carbon dioxide levels, notes UNCC:Learn",
        "understanding of environmental issues e.g. Recognizing
           that burning fossil fuels for power, cutting down
           forests, and manufacturing goods are major causes of
           climate change and that these actions lead to effects
           like rising temperatures and extreme weather events,
           according to the United Nations and the National
           Oceanic and Atmospheric Administration (NOAA).",
        "climate literacy and awareness, e.g. analyzing news
           articles or social media posts about climate change,
           discerning scientific consensus from misinformation,
           and understanding the uncertainties in scientific
           findings"
    ],
    "Affective": [
        "emotional response to climate change, e.g. This category
           includes a range of emotions such as eco-anxiety, a
           chronic fear of environmental doom; guilt about
           personal lifestyle choices; sadness and grief over the
           loss of ecosystems; and anger at inaction from
           governments and corporations. These negative emotions
           can be contrasted with hope and optimism, which can
           also motivate positive action.",
        "motivation to take action, e.g. The motivation to act on
           climate change is driven by a sense of duty and
           responsibility to future generations. It also relies on
            a person's efficacy beliefs, or the conviction that
           their actions can make a difference. Health and safety
           concerns, when climate change is framed as a direct
           threat to personal well-being, are also powerful
           motivators.",
        "concern and care for the environment, e.g. A person's
           level of concern is strongly linked to their emotional
           connection to nature and a deep sense of belonging to
           the natural world. Conversely, psychological distancing
           ---the feeling that climate change is a problem for
           someone else---can reduce a person's concern. Finally,
           social norms and influence play a role, as people are
           often motivated to adopt pro-environmental behaviors to
            align with their social group. "
    ],
    "Behavioral": [
```

```
        "Actions to mitigate climate change: These encompass both individual and large-scale actions aimed at reducing or preventing greenhouse gas emissions. Examples include reducing personal energy consumption, adopting renewable energy sources, and switching to sustainable transport like walking, biking, or electric vehicles. It also involves lifestyle changes such as eating more plant-based foods, reducing food waste, and consciously minimizing consumption of goods like fast fashion and plastics.",
        "Community participation in sustainability: This involves the active and genuine involvement of people in addressing local problems and implementing projects to promote sustainable development. Participation can be driven by a motivation to improve one's community, building a sense of collective purpose. Strategies include community needs assessments, mobilization, and working with local authorities. It is crucial that participation is inclusive, involving women and other vulnerable groups, and that agencies or governments respect local knowledge and needs.",
        "Adoption of eco-friendly practices: The adoption of eco-friendly practices by individuals and organizations includes a range of actions that minimize environmental harm, reduce waste, and conserve energy. Examples involve using reusable items, recycling, upcycling, composting, and conserving water. In the corporate sector, this includes implementing strategies to reduce a company's carbon footprint, investing in sustainable supply chains, and offering eco-friendly products."
        ],
    "Institutional": [
        "policy development for climate adaptation: Effective policy-making for climate change adaptation involves national and sub-national governments developing strategies and measures to reduce vulnerability to climate impacts. This includes implementing large-scale measures like enhancing early warning systems, strengthening infrastructure, and developing strategies for managing water, land, and food systems. It also requires a 'whole-of-government' approach and integrating climate considerations into development plans.",
        "Governance and institutional support: This involves building robust institutional arrangements and governance models to address social problems like climate change and ensure sustainable development. Good governance is characterized by participatory behaviour, transparency, accountability, and coordination across different levels of government. Institutional support, often technical and strategic, is aimed at
```

```
            strengthening government departments to improve
            financial management, service delivery, and overall
            effectiveness.",
        "Capacity building in organizations: This is the process
            of helping individuals, organizations, and communities
            strengthen their skills, knowledge, and systems to
            achieve their goals. For organizations, this involves
            upskilling and reskilling employees, developing
            leadership, forming partnerships, and upgrading
            technology and processes. Ultimately, capacity building
             aims to transform behaviours and improve an
            organization's ability to achieve its mission."
    ]
}
```

2 Appendix B: A list of queries used for querying data

```

queries = """
              tree planting initiatives South Africa,
              coastal conservation South Africa,
              wetland restoration South Africa,

              climate change education South Africa,
              environmental curriculum development South Africa,
              climate change workshops South Africa,
              sustainability training South Africa,
              climate change awareness campaigns South Africa,
              community climate resilience South Africa,
              environmental activism South Africa
         """
```

3 Appendix C: Data extraction prompts

```

data_extraction_prompt_for_interventions = """
You are an AI assistant specializing in identifying and classifying content related to **Sustainable Development Goal 13 (Climate Action)** --- specifically targets 13.3.
Your task is to follow the instructions as listed below:

If the text **does not reference any climate change-related concepts or ideas **, mark it as **NOT RELATED TO CLIMATE CHANGE**.

---

Document:
<document>
{document}
</document>

Chunk to analyze:
<chunk>
{chunk}
</chunk>

---

Instructions:
1. Read the entire document carefully.
2. Produce a detailed summary organised under the following headings:
   - Context & Setting: Where the work takes place (region e.g. (Mpumalanga, uMkhanyakude District Municipality), communities, ecosystems, timeframe).
   - Specific country: The country where this work has taken place (e.g., South Africa, India, USA).
   - Population / Stakeholders: Who is affected or involved (e.g., local communities, schools, government agencies).
   - Interventions or Actions: Specific climate-related interventions such as:
     Mitigation (e.g., reducing greenhouse gas emissions),
     Adaptation (e.g., preparing for climate impacts),
     Ecosystem-based solutions (tree planting, wetland restoration, coastal conservation),
     or Awareness & Education Efforts.
   - Outcomes & Impacts: Quantitative or qualitative impacts--- include numbers if mentioned (e.g., number of trees planted, training sessions held, policy changes).
   - Links to SDG Targets: Explicit or implied alignment with SDG 13.3.
3. Write the summary as neutral factual prose, NOT bullet points.
```

```
4. For dates, Paraphrase for clarity but do not invent dates or targets.
5. If no specific date is mentioned, indicate it as "Not specified"

{{
  "summary": "<Use the detailed summary to populate this section or if not related to climate change output 'NOT RELATED TO CLIMATE CHANGE'>",
  "community": "<Who or what is affected? e.g., South African communities, ecosystems, schools>",
  "project_name: "<What is the name of the project? e.g. Green Roots Initiative, Trees for Tomorrow, Urban Canopy Project, Blue Coast Guardians, Wetlands for Wildlife>",
  "time_frame": "<When the project started, when it is planned to end, or any deadlines mentioned.>",
  "future_goals": "<Any objectives, targets, or milestones the project aims to achieve in the future.>",
  "intervention": "<What action / intervention is described? e.g., tree planting, workshops, curriculum development>",
  "outcome": "<What is achieved or changed? e.g., increased awareness, restored wetlands, reduced emissions.>",
  "impact": "<Impacts explicitly mentioned in the text --- can be numerical (e.g., '5000 trees planted') or qualitative (e.g., 'improved community knowledge.')>",
  "sdg_13_3":"<If the intervention aligns with SDG 13.3, answer YES or NO and add a reason.>",
  "sa_project": "<If this is a South African initiative output YES otherwise NO.>",
  "climate_related": "<If this is a climate change related, output YES otherwise NO.>"
}}

Only return the JSON object --- no explanation, no extra commentary.

"""
```

4 Appendix D: Code and prompts for generating semantic similarity and sentiment analysis heat map

```
import json
from tqdm import tqdm

def llm_judge_sentiment_temporal(text, field_name):
    if not text or text.strip().lower() in ["not specified", "none"]:
      return {"sentiment": "neutral", "timeframe": "present"}

    prompt = f"""
    You are analyzing text extracted from a climate-related research article.
    The text is from the **{field_name}** section.

    Determine:
    1. Sentiment polarity: "positive", "negative", or "neutral"
    2. Temporal context: "past", "present", or "future"

    Respond strictly in JSON format, for example:
    {{
      "sentiment": "positive",
      "timeframe": "future"
    }}

    Text: "{text}"
    """

    try:
        messages = [{"role": "user", "content": prompt}]
        response = client.invoke(messages)
        return response.content.strip()
    except Exception as e:
        print(f"Error for {field_name}: {e}")
        return {"sentiment": "neutral", "timeframe": "present"}

def safe_parse_response(response):
    """Ensure the LLM result is a dict with 'sentiment' and 'timeframe'."""
    if isinstance(response, dict):
        return response
    if isinstance(response, str):
        try:
            # try parsing JSON string
            return json.loads(response)
        except Exception:
            pass
    # fallback in case of error
    return {"sentiment": "neutral", "timeframe": "unspecified"}
```

```python
fields = ["interventions", "outcomes", "impacts", "time_frames", "
    future_goals"]
error_data = []
for field in tqdm(fields):
  try:

    print(f"Processing {field}...")
    results = new_data_df[field].apply(
        lambda x: safe_parse_response(llm_judge_sentiment_temporal
           (str(x), field))
    )

    new_data_df[f"{field}_sentiment"] = results.apply(lambda r: r.
       get("sentiment", "neutral"))
    new_data_df[f"{field}_timeframe"] = results.apply(lambda r: r.
       get("timeframe", "unspecified"))

  except Exception as e:
    error_data.append({"field": field, "error": str(e), "results":
        results})
    print(f"Error for {field}: {e}, results: {results}")
    continue
```